Vijay Chevireddi

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Education

Master of Engineering, Robotics — University of Maryland (UMD), *College Park, MD* — Aug 2023 – May 2025
Bachelor of Engineering — Osmania University, *India* — Aug 2018 – June 2022

Skills

- **Programming:** Python, Pytest, OpenCV, SQL (Postgres), OpenSearch, Pgvector, FastAPI, CUDA, Linux
- **Deep Learning Frameworks:** PyTorch , TensorFlow, Hugging Face , TensorRT, Scikit-learn , XGBoost , Spark, JAX
- **ML-Ops:** Docker, Torchserve, RAG, Chains/Agents using LangChain/LangGraph, Tool binding, Kubernetes deployment, Airflow, DAGs execution, Design Patterns, Terraform for IaC (Infrastructure as Code), AWS, SageMaker, Git, CI/CD, data pipelines, LanceDB, Qdrant
- **AI:** Machine learning, Deep Learning, Neural Networks, Computer Vision, NLP, Multi-modal Transformers, Large Language Models, Vision Transformers, CNNs, Object Detection (e.g., YOLO), NLP, Generative Models (core concepts), Fine-Tuning, AI Alignment
- **Robotics:** Perception, localization, Mapping, ROS2, C++, Solidworks, Gazebo, Point Cloud data, NVIDIA Jetson Orin

Certifications

- AWS Certified Machine Learning - Specialty
- AWS Certified Cloud Practitioner

Experience

Machine Learning Research Engineer, *Fischell Department of Bioengineering – UMD* — June 2024 – May 2025

- Developed and refined a real-time object detection and tracking pipeline for sonar imagery by training and fine-tuning various **YOLOv8** models. The YOLOv8n model performed best, achieving a test accuracy of **90%**.
- Improved pipeline performance by **20%** by upgrading the architecture to a **Real-Time Detection Transformer (RT-DETR)** and integrating **BYTETracker** with custom post-processing logic.
- Engineered a human gait classification pipeline by extracting 17-point skeletons using the MMPOSE framework and implementing a normalization workflow to create scale-invariant features.
- Achieved **95%** accuracy on the test set by training **vision transformer, RNN, and CNN-LSTM** models on temporal skeletal data to classify gait patterns.

Machine Learning Engineer, *Sai Vamsi Industries – India* — Jan 2022 – May 2023

- Constructed and deployed an automated visual inspection system by fine-tuning a **YOLO** model on a custom dataset to accurately detect and classify anomalies in machine components.
- Executed a comparative analysis of various deep learning models, including **ResNet, VGG, and Vision Transformer (ViT)**, to optimize classification performance.
- Determined the Vision Transformer (ViT) as the most effective model, which yielded a top test accuracy of **93%**.
- Integrated a sophisticated reporting module using the **CLIP captioning** model and the **ChatGPT API** to perform programmatic root cause analysis, leveraging few-shot learning to ensure high-quality context-aware reports.

Projects

Advanced Vision Systems for Real-World Autonomous Navigation (Competition Winner)

- Developed and integrated a highly-specialized traffic sign recognition module by **fine-tuning a YOLOv8 model** on a custom dataset.
- Engineered a multi-faceted perception system where **homography** was used for dynamic route planning on the road surface, while **optical flow** detected and calculated the precise velocity of moving obstacles for proactive collision avoidance.
- Established robust spatial awareness by applying projective geometry to identify the **horizon and vanishing points**, which provided critical information on obstacle locations relative to the robot. Successfully deployed the entire perception pipeline from simulation using **ROS2 and Gazebo** to a **physical robot**, achieving over **80%** navigation success in unfamiliar real-world terrains.

Transformer Based NLP Pipeline for Robot Navigation with LoRA Optimization

- Built a natural language interface using a **teacher-student learning** framework, **fine-tuning** a **T5-Small "student" model** in **PyTorch** with **Low-Rank Adaptation (LoRA)** on a dataset generated by the **GPT-4 "teacher" model**.
- Applied Low-Rank Adaptation (LoRA) to reduce training parameters by **85%** while retaining **92.5%** test accuracy.
- Operationalized the resulting **NLP** model with **ROS2 and Gazebo** to validate autonomous navigation on a TurtleBot3.

Vision-Augmented Deep-Q Learning for Mario Using SWIN Transformer

- Conceptualized an advanced **agent** for Super Mario Bros using a **Double Deep Q-Network (DDQN)** with a novel hybrid architecture that integrates a pretrained **ResNet-18** and a **Swin Transformer**, significantly improving learning efficiency and gameplay strategy.
- Achieved superior and more stable model performance, increasing the 500-episode moving average reward from approximately 2,200 to nearly 2,700. This surpassed basic CNN models that exhibited significant learning fluctuations.
- Leveraged **Docker** to containerize the project's environment and dependencies, ensuring both reproducibility and a simplified setup.

Transformers for 3D Object Detection in LiDAR Point Clouds

- Designed and trained a custom transformer-based **3D object detection model** for LiDAR point clouds on the KITTI dataset.
- Fused PointNet++ embeddings with specialized transformer encoders and introduced a novel loss function, achieving a 3D bounding box detection mAP IoU of 0.67—on par with leading benchmarks.